Heatwave Prediction in Delhi Using Meteorological Time-Series Data: An Exploratory Data Analysis and Literature Review

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Abstract—This paper presents an exploratory data analysis (EDA) of hourly meteorological observations for 31 Indian cities (Jan 2023–Jan 2024), focusing on Delhi (28.6139°N, 77.209°E). We characterise the heatwave binary label and its relationship with 15 meteorological predictors. Key findings include a class imbalance of ≈14.4 %, strong feature separability via maximum temperature (Spearman $\rho \approx 0.51$) and dew point (Standardized Mean Difference (SMD) = +0.94), and ≈10–11 % missing data across most variables. The review contextualises these findings within existing literature on heatwave detection, climate machine learning, and Indian urban meteorology.

Index Terms—heatwave prediction, exploratory data analysis, Indian climate, class imbalance, time-series classification, urban meteorology.

I. INTRODUCTION

Heatwaves are among the most lethal natural hazards globally, and their frequency and intensity are rising due to anthropogenic climate change [1]. Delhi, located in the Indo-Gangetic plain, regularly records maximum temperatures exceeding 45 °C during the pre-monsoon season. The India Meteorological Department (IMD) defines a heatwave when the maximum temperature reaches at least 40 °C and departs from normal by ≥4.5 °C over at least two consecutive days [2].

Machine learning (ML) methods have emerged as powerful complements to numerical weather prediction (NWP) for short-range heatwave forecasting. Recent data-driven models such as GraphCast [3] have demonstrated that learned atmospheric representations can match or exceed traditional NWP skill, motivating ML-based approaches at the city scale. A rigorous EDA is a prerequisite for any ML pipeline: quantifying missing data, verifying label integrity, understanding seasonality, and identifying predictive features are critical steps—especially under class imbalance and temporal autocorrelation.

This paper reports an EDA of the india_weather_data.csv dataset comprising 131,055 hourly records across 31 coordinate pairs corresponding to Indian monitoring stations. After filtering to Delhi, the working subset is 9,505 rows with a heatwave label prevalence of 14.35 %.

II. RELATED WORK

A. Heatwave Detection and Climatological Definitions

Perkins and Alexander [4] reviewed 16 heatwave definitions and concluded that metrics combining both temperature and duration captured societal impact better than single-threshold indices. Rohini et al. [5] analysed IMD station data (1951–2014) and identified increasing heatwave trends over central and northern India, with maximum temperature and humidity jointly determining heat stress. Critically, both IMD [2] and World Meteorological Organization (WMO) guidelines require multi-day persistence (typically 2+ days) as a necessary condition for heatwave classification—a threshold that bears directly on the label validity discussion in Section IV.

Beyond temperature thresholds, Raymond et al. [6] demonstrated that wet-bulb temperatures exceeding 35 °C represent a human survivability limit, motivating inclusion of humidity-derived features (dew point, heat index) as first-class predictors alongside dry-bulb maximum temperature.

B. Machine Learning for Extreme Heat and NWP Post-Processing

Chattopadhyay et al. [7] applied convolutional neural networks to reanalysis data, showing that spatial feature extraction substantially outperformed point-based logistic regression for extreme weather pattern prediction. Rasp and Lerch [8] demonstrated that neural-network post-processing of NWP ensembles reduced temperature forecast error by up to 30 % over traditional model output statistics. For tabular station-level data, gradient-boosted tree ensembles consistently achieve competitive F1 scores on meteorological classification tasks, making them a natural baseline for our dataset.

C. Class Imbalance and Evaluation Metrics

Extreme weather events are inherently rare, producing heavily imbalanced labels. Branco et al. [9] found that cost-sensitive methods and synthetic oversampling (SMOTE) consistently outperform accuracy-optimised classifiers for rare-event detection. Precision–Recall Area Under the Curve (PR-AUC) is recommended over Receiver Operating Characteristic Area Under the Curve (ROC-AUC) when positive-class prevalence falls below 20 %—a threshold directly applicable to our dataset's 14.4 % heatwave rate.

D. Missing Data and Feature Engineering

Little and Rubin [10] emphasised that naive mean imputation distorts time-series autocorrelation; forward-fill followed by seasonal-median imputation better preserves temporal structure. Calendar-based features (month, hour-of-day) and lag features—rolling means over 3–24 h windows—are strong predictors in urban temperature models but must be constructed exclusively from past observations to prevent data leakage in time-series cross-validation.

III. DATASET DESCRIPTION AND KEY EDA FINDINGS

The dataset contains 131,055 rows and 18 columns of hourly meteorological variables for 31 Indian monitoring stations (Jan 2023–Jan 2024), identified by (latitude, longitude) coordinate pairs which may not correspond to 31 distinct administrative cities. Variables include wind_speed, cloud_cover, dew_point, precipitation_probability, pressure_surface_level, uv_index, heatwave (binary target), visibility, rainfall, solar_radiation, max_temperature, min_temperature, max_humidity, and min_humidity.

After filtering to Delhi and removing invalid timestamps, the working subset is 9,505 rows. Of these, 2,431 rows are exact duplicate (date, latitude, longitude) triplets, likely arising from a data-merging artifact when combining multiple source feeds; the first occurrence will be retained during preprocessing. The heatwave label has 1,224 positive, 7,303 negative, and 978 missing observations (prevalence 14.35 %).

Feature Separability: Standardized Mean Difference (SMD) analysis identifies min_temperature (SMD = +1.32), max_temperature (SMD = +1.18), and dew_point (SMD = +0.94) as the top discriminating features. The Spearman correlation of max_temperature with heatwave is $\rho \approx 0.51$ ($p < 0.001$) — a moderate association in predictive

modeling terms that nonetheless confirms temperature as the primary signal. Humidity variables (min_humidity SMD = −0.83) confirm the dry-heat paradigm of north Indian summers [2].

Seasonality: Monthly heatwave rates peak in May (17.7 %) and unexpectedly in August (16.6 %). The August peak is climatologically anomalous—Delhi's documented heatwave season is overwhelmingly pre-monsoon (April–June) [5]—and may reflect the label quality issue discussed below rather than a true meteorological signal. Missingness: All feature columns show ≈10–11 % missing rates, consistent with periodic sensor maintenance rather than weather-event-correlated outages.

Label Validity Caveat: Streak analysis identifies 1,130 heatwave segments post-deduplication, with a median duration of 1 h and a maximum of 3 consecutive hours. This is fundamentally inconsistent with any standard operational definition—IMD, WMO, and Perkins & Alexander [4] all require multi-day persistence. The label most likely encodes instantaneous heat-stress exceedance rather than a canonical heatwave. Downstream models trained on this label will therefore predict heat-stress episodes, not true meteorological heatwaves; results should be interpreted accordingly, and relabelling using a persistence-based criterion is recommended before operational deployment.

IV. COMPARATIVE ANALYSIS AND MODELING IMPLICATIONS

The EDA results yield four direct implications. First, the 14.4 % positive-class prevalence necessitates class-weighted loss functions or SMOTE oversampling, with PR-AUC and macro-F1 as primary evaluation metrics [9]. Second, 2,431 duplicate timestamps must be removed (keep-first strategy) before feature engineering to avoid erroneously short lag windows. Third, time-based splitting (training on 2023-01–2023-09, testing on 2023-10–2024-01) is mandatory to prevent leakage. Fourth, forward-fill imputation for gaps ≤2 h followed by seasonal-median imputation for longer gaps is appropriate given the MAR-consistent missingness [10].

The moderate separability of max_temperature ($\rho \approx 0.51$) suggests a linear classifier with threshold tuning may achieve respectable recall as a baseline. The interaction between dew point, cloud cover, and wet-bulb-derived features [6] may benefit from non-linear gradient-boosted trees. The anomalous August heatwave rate and single-hour streaks reinforce the recommendation to validate and potentially reconstruct the target label before committing to a final model architecture. The findings are consistent with prior literature in identifying temperature and humidity as primary drivers of heatwave conditions.

V. CONCLUSION

This paper presents a systematic EDA of hourly weather data for Delhi in the context of binary heatwave prediction. Significant class imbalance (14.4 %), moderate feature separability (max_temperature $\rho \approx 0.51$), ≈10 % uniform missingness, and critically short heatwave streaks (median 1 h) inconsistent with standard multi-day definitions were identified. The label validity issue is the most important finding: future work should reconstruct the target using a persistence-based IMD-compliant criterion before model training. Conditional on label quality, time-aware gradient-boosted classifiers with humidity-derived and lag features, evaluated using PR-AUC and macro-F1, are recommended as the modeling strategy.

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